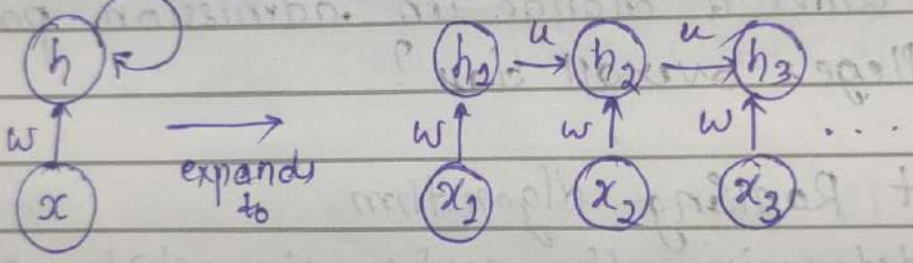
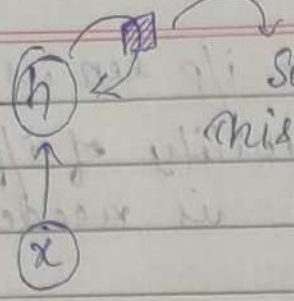
RNN Recurrent Neural Network

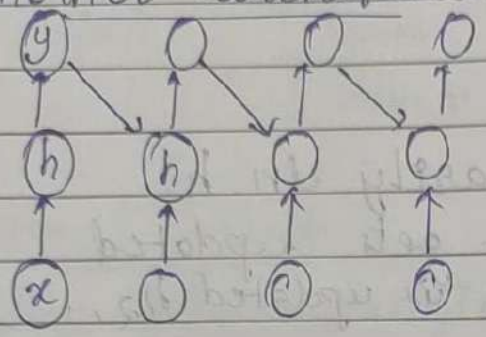
→ i/p data : Sentences
↳ varying length of sequence



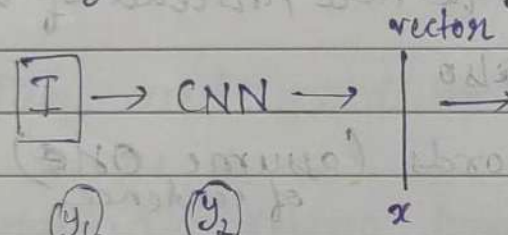


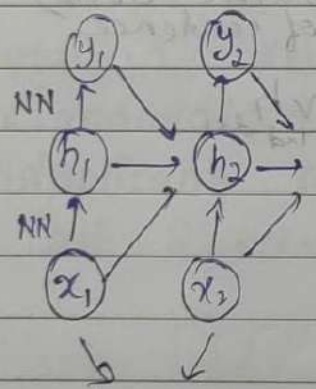
Some texts show this
This shows time delay
$(h_{t-1} \rightarrow h_t)$

* Another architecture



Q) Design an NN to generate caption of an image





This must happen such that
v token

Each component of x.

$u, w \rightarrow$ fixed

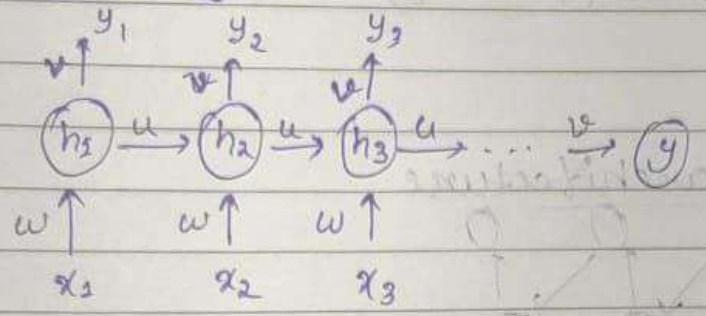
$h_1 = wx_1$

$h_2 = wx_2 + h_1 \cdot u$

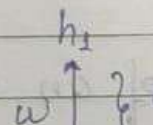
$h_3 = wx_2 + u h_2$

Only 1 w but i/p len is not fixed.

→ Flexibility of i/p size is needed.



→ Info is stored primarily in h.
It is one vector that gets updated
(h_2 is updated h_1, h_3 is updated h_2, ...)



} This can be a network here, instead of W
similarly u & v also

The $x_1, x_2, \ldots$ may be words (assume OHE of sentence)

$x \in \mathbb{R}^v$ $\quad w \in \mathbb{R}^{d \times v}$

$h \in \mathbb{R}^d$ $\quad v \in \mathbb{R}^{1 \times d}$

$y \in \mathbb{R}$

$\underset{1\times1}{y} = \underset{1\times d}{v} \underset{d\times1}{h}$

Part of Speech Tagging

→ output is required from each x

$x_i \rightarrow h_i \rightarrow y_i$ $\quad \therefore y_i$ is POS of x_i

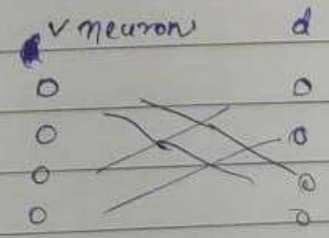
Sentiment Analysis

→ output y is enough

$h^t = f(w, h^{t-1}, x)$

→ Hence Recurrent.

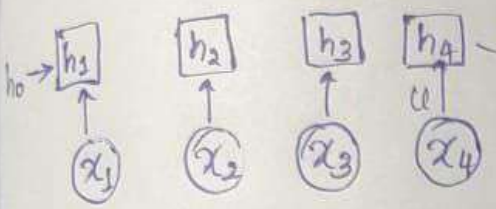
$x \rightarrow h$ can happen like



* Consider a simple model $h_t = Ux_t + Vh_{t-1}$

$$c = Ux_T + h_{T-1}$$
$$= Ux_T + V(ux_{T-1} + vh_{T-2})$$

* Assume only 4 inputs



$\xrightarrow{v}$ $c = Ux_4 + V(ux_3 + V(ux_2 + V(ux_1 + Vh_0)))$

can be used if required.

$\therefore c = Ux_4 + vux_3 + v^2ux_2 + v^3ux_1 + v^4h_0$

Dependency of c on $x_1 \propto v^3u$ ⇒ this is a very small value

c on $x_4 \propto u$

$\therefore$ Capturing long term dependencies is a challenge.

If $u, v > 1$ ⇒ Explode $u, v < 1$ ⇒ vanish

Training RNN

→ To learn u and v

→ At c, we have ground truth ⇒ compute loss

$E = (t - c)^2$

$\frac{\partial E}{\partial u}$ and $\frac{\partial E}{\partial v}$

$$\therefore \frac{\partial E}{\partial u} = \frac{\partial E}{\partial h_4}\frac{\partial h_4}{\partial u} + \frac{\partial E}{\partial h_4}\frac{\partial h_4}{\partial h_3}\frac{\partial h_3}{\partial u} + \frac{\partial E}{\partial h_4}\frac{\partial h_4}{\partial h_3}\frac{\partial h_3}{\partial h_2}\frac{\partial h_2}{\partial u} + \frac{\partial E}{\partial h_1}\frac{\partial h_1}{\partial u}$$

$$\frac{\partial E}{\partial v} = \frac{\partial E}{\partial h_4}\frac{\partial h_4}{\partial v} + \frac{\partial E}{\partial h_4}\frac{\partial h_3}{\partial v} + \overset{\text{for } (h_2)}{(\ )} + \overset{\text{for } (h_1)}{(\ )}$$

$$= \sum_{t=1}^{T} \frac{\partial E}{\partial h_t}\frac{\partial h_t}{\partial v}$$

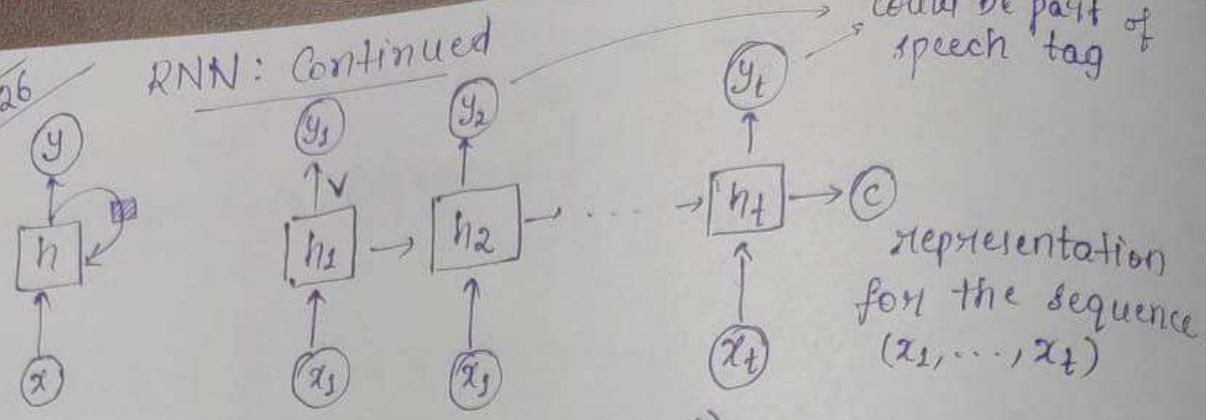
"Back Propagation through Time"

$$= \sum_{t=1}^{T} \frac{\partial E}{\partial h_t}\frac{\partial h_t}{\partial u}$$

As $t \uparrow$, number of gradients multiplied $\uparrow$ ⇒ Since gradients are typically $0 < \ < 1$, product is $\downarrow\downarrow$. o/w > 1, product $\uparrow\uparrow$ ⇒ Vanishing gradient / Exploding gradient

19/2/26

RNN: Continued

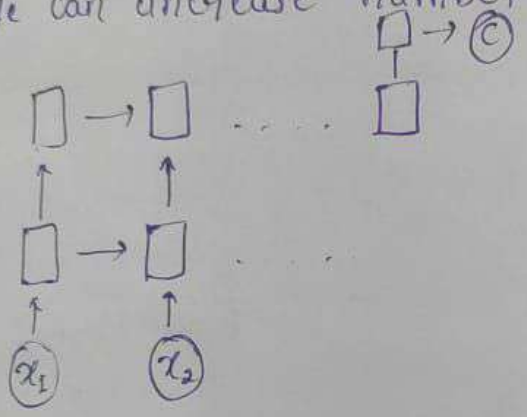


$y_1 = \text{argmax}(\text{softmax}(v(h_1)))$

or

$y_1 =$ " (" $(f(h_1)))$

→ We can increase number of hidden layers.

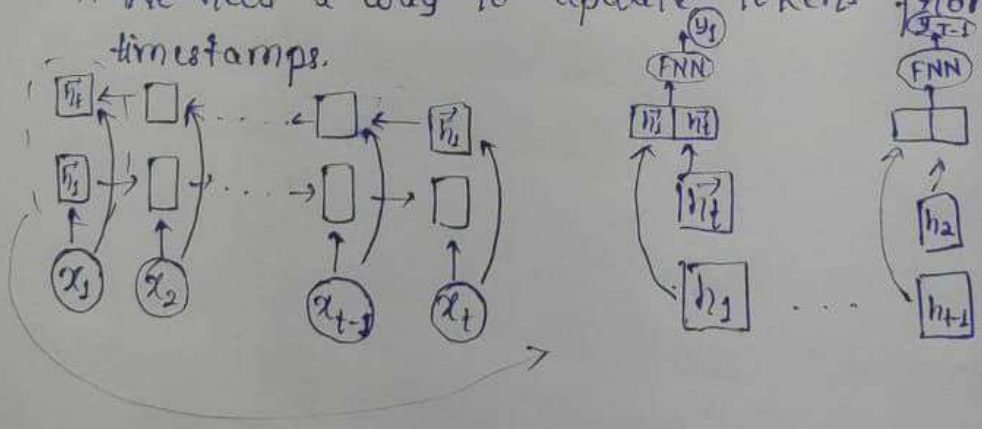


RNN can be used for
- text / speech tasks
  - classification / recognition.

eg: Automatic grammar correction

I have a apple : change a → an

∴ We need a way to update tokens from previous timestamps.



"Bidirectional RNN" : only hidden states go in both directions. Not input.

How to stack LSTMs?

→ Make h_t go to another LSTM block

Schedule : Monday: RNN, LSTM lab

Tuesday: finish RNN, LSTM

Thursday: CT3

Monday: Mid term Lab (1 side A4 cheatsheet)

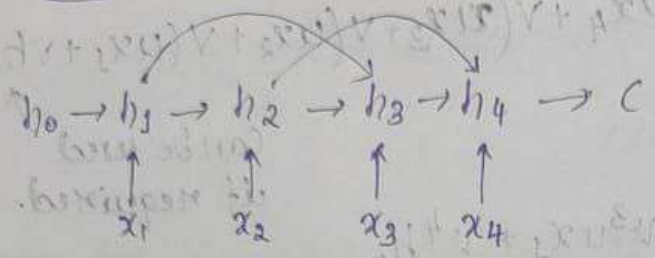
Next Monday: Midterm project evaluation.

→ Both are bad.

* $V = S \Lambda S^T$ → $V^k = S \Lambda^k S^T$ Eigenvalues.
  $V^2 = S \Lambda^2 S^T$

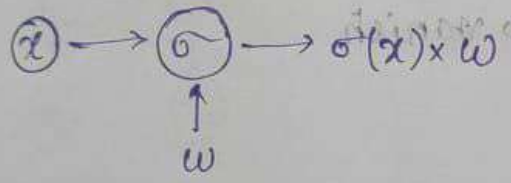
  As $k\uparrow$, the elements in $V \downarrow$.

→ One strategy

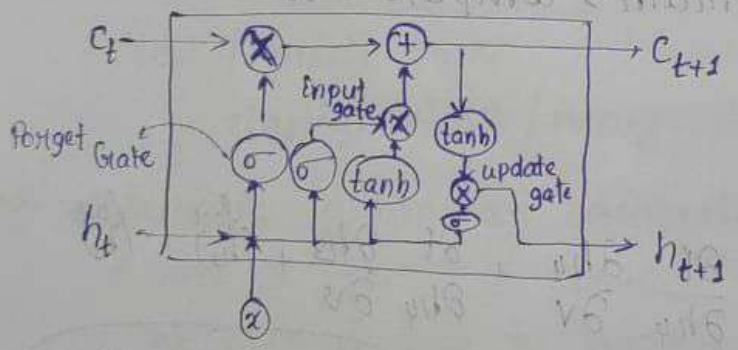


LONG SHORT TERM MEMORY NETWORK

→ Gates control which information flows.



$\sigma(x) \times w$ → Control how much w goes through.



→ All gates are sigmoid

Forget gate ~~$\sigma([h,x]$~~ $\sigma(w_f[h,x])$ f

Input gate $\sigma(w_i[h,x])$ i

Update gate $\sigma(w_u[h,x])$ u

$c_{t+1} = f \bullet c_t + i \bullet \tanh([h_t, x_t])$

$h_{t+1} = \tanh(c_{t+1}) \cdot u$

★ Kind of SOTA until transformer came in 2020

★ Google translate used it